

Aircraft Aeroacoustics 0860395

Project 2 - Aircraft noise analysis

December 14, 2025

In this project, you will engage with aircraft noise analysis, gaining an understanding of compact noise sources, the effect of motion on acoustic sources, and cavity airframe noise. There are three sections to be answered in this project, as presented below. In part A (30 points of the project grade), you will deal with compact noise sources and learn about their significance when assessing the noise sources of aerial systems. In part B (35 points of the project grade), you will be introduced to the effect of motion on acoustic sources and learn about the Doppler effect as well as Green's function. These elements are essential for analyzing realistic scenarios of a moving acoustic source relative to an observer, as we will see later throughout the course. Finally, in part C (35 points of the project grade), you will learn about airframe cavity noise, develop the classical Rossiter's semi-empirical model, and utilize it to analyze the acoustics of an approaching aircraft. A total of 20 bonus points is available in this project (where a bolded **Bonus** text is presented); thus, the maximum grade available for this project is 120.

Part A - Compact Noise Sources

30 points (+ 5 bonus points)

In the class, we developed the governing conservation equations for linear acoustics as well as the homogeneous wave equation in Cartesian coordinates (x_1, x_2, x_3) and in spherical coordinates (r, θ, φ) :

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = 0 \quad \text{or} \quad \frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{1}{r} \frac{\partial^2}{\partial r^2} (rp') = 0, \quad (1)$$

where, for spherical coordinates, we assume uniform radial propagation. The speed of sound is denoted by c_0 . Assuming a harmonic sound signal of the form $p'(r, t) = \mathcal{R} \{ \hat{p}(r) e^{-i\omega t} \}$, one can also express the wave equation in the frequency domain as follows:

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r\hat{p}) + k^2 \hat{p} = 0, \quad (2)$$

where $\hat{p}(r) = \hat{A}e^{ikr}/r$ is the complex amplitude of the pressure perturbation signal, $k = \omega/c_o$ is the acoustic wavenumber, ω is the angular frequency, and $\hat{A} = Ae^{i\phi}$ takes into account the phase. Under the assumption of acoustically compact bodies ($ka \ll 1$), where a is a characteristic body length, we found the following solutions for a monopole, dipole, and a lateral quadrupole (see Fig. 1):

$$\begin{aligned} [\hat{p}(r)]_{\text{monopole}} &= -i\omega\rho_o\hat{Q}\frac{e^{ikr}}{4\pi r} \\ [\hat{p}(r)]_{\text{dipole}} &= -ik\cos\theta e^{ikr}\frac{\hat{F}_1}{4\pi r}\left(1 - \frac{1}{ikr}\right) \\ [\hat{p}(r)]_{\text{quadrupole}} &= \frac{2(ka)^2}{9}\left(\frac{x_1x_2}{r^2}\right)\left(\frac{i\omega\rho_ov_oa^2e^{ikr}}{r}\right)\left(1 + \frac{3i}{kr} - \frac{3}{(kr)^2}\right), \end{aligned} \quad (3)$$

where ρ_o is the mean density in the medium, $\hat{Q} = 4\pi v_o a^2$ is the complex amplitude of the volumetric flow rate, and the complex amplitude of the force exerted on the body in the x_1 direction is given by $\hat{F}_1 = -2\pi i\omega\rho_ov_oa^3$. The complex amplitude of the body's surface motion is given by v_o (e.g., pulsating boundary, translation motion, etc.).

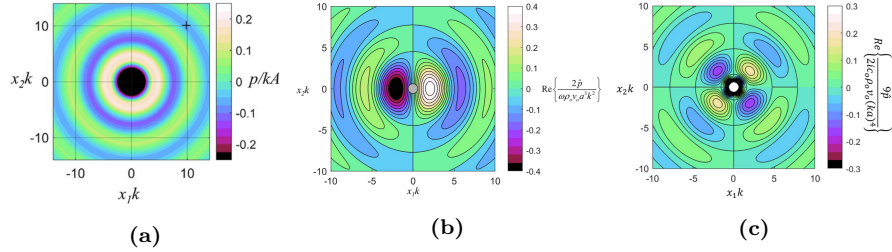


Figure 1: Sound pressure fields of (a) monopole, (b) dipole, and (c) lateral quadrupole (in the far-field, $kr \gg 1$) sources [1].

- **A.1** Based on the elementary noise sources defined in Eq. 3, identify the dominant compact source type (monopole/dipole/quadrupole) associated with the **noise mechanisms below** in low-Mach number conditions. Please justify your selection based on physical reasoning of the flow-induced noise scenario and our interpretation of the elementary noise sources. **(4 points)**

1. Acoustic narrow-band tone at the blade-passing frequency (f_b) from a propeller/rotor due to periodic aerodynamic loading.
2. Broadband turbulent jet mixing noise from a subsonic single-stream jet.
3. Noise from vortex shedding over a landing-gear strut.
4. Noise from a cabin outflow/bleed vent (unsteady mass flow through a hole).

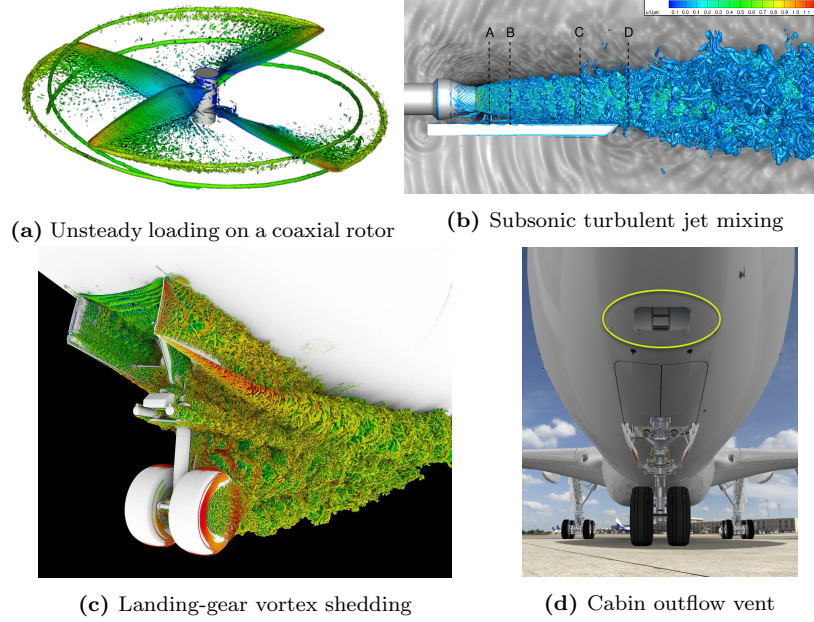


Figure 2: Examples of common noise mechanisms in aeronautics.

Commonly, linear acoustics are described up to the idea that real aeroacoustic sources can be represented as **superpositions** of compact monopoles, dipoles, and quadrupoles, particularly when the length scale of the system of sources d is small compared with the acoustic wavelength λ . Using elementary monopole sources ($\hat{p}(r) = i\omega\rho_o\hat{Q}e^{ikr}/4\pi r$), the superposition principle, and the assumption of acoustically compact source systems ($kd \ll 1$), we found in class the following approximations for a dipole and a lateral quadrupole:

$$\begin{aligned}
 [\hat{p}(r)]_{\text{dipole}} &\approx ikd \cos\theta \left(\frac{i\omega\rho_o\hat{Q}e^{ikr}}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right) \\
 [\hat{p}(r)]_{\text{quadrupole}} &\approx (kd)^2 \left(\frac{x_1x_2}{r^2} \right) \left(\frac{i\omega\rho_o\hat{Q}e^{ikr}}{4\pi r} \right) \left(1 + \frac{3i}{kr} - \frac{3}{(kr)^2} \right)
 \end{aligned} \tag{4}$$

Use the principle of superposition to answer the following question.

- **A.2** Two harmonic monopole sources lie on the x_1 axis at position $x_1 = 0$ and $x_1 = d$. Given that the source at the origin $(x_1, x_2) = (0, 0)$ has a complex strength $\hat{Q}$ (i.e., complex amplitude of its volumetric flow rate):
 1. Calculate the strength of the source at $x_1 = d$, necessary to produce zero sound pressure at a point on the x_2 axis given by $x_2 = R$. **(2 points)**
 2. Show how the results may be approximated when $R \gg d$ and $\lambda \gg d$, both in the near- and far-fields. **(2 points)**

3. In addition to the source at $x_1 = 0$ (with $\hat{Q}$) and $x_1 = d$ (with $\hat{Q}_1$), a third source is now added at $x_1 = -d$ (with $\hat{Q}_2$). Calculate the strength of the sources at $x_1 = d$ and $x_1 = -d$ necessary to produce zero sound pressure at both $x_2 = R$ and $x_2 = -R$. **(2 points)**

A key aspect in analyzing aeroacoustic noise sources is their compactness and characteristics in the near- and far-fields. In the following two questions, you will gain a better understanding of these important aspects.

- **A.3** Derive the sound field produced by a translating sphere of radius a , but without using the compact approximation. Factorize your answer into terms representing the far-field sound, non-compact, and near-field effects. Use this factorization to plot the error in dB in the compact approximation, as a function of ka , and in the far-field approximation as a function of kr . For what ranges of ka and kr are these approximations accurate to within 1 dB and 1% (0.09 dB)? **(10 points)**
- **A.4** An aircraft approaching to land (with its engine on idle) generates tonal noise recorded by a far-field observer located on the ground at a radial distance of 90 m from the air vehicle. The vehicle generates two narrow-band tones, one with an SPL of 63 dB(re 20 μ Pa) at $f_1 = 500$ Hz, and the second with an SPL of 59 dB(re 20 μ Pa) at $f_2 = 2000$ Hz. The speed of sound is $c_o = 343$ m/s, the air density is $\rho_o = 1.225$ kg/m³, and the atmospheric pressure is $p_o = 101,325$ Pa. In your answer, neglect the Doppler effect due to the motion of the aircraft relative to the observer.
 1. What is the SPL of the overall sound heard by the observer? **(3 points)**
 2. How small must the noise-generating device be to form a compact source at 500 Hz and at 2000 Hz? **(2 points)**
 3. From inspecting the directivity of the noise spectra, you determined that the two noise sources at f_1 and f_2 are compact monopoles. Estimate the volume velocity amplitude $\hat{Q}$ of each of those sources. State the units of your answer. **(5 points)**
 4. **Bonus** Inspection of the aircraft logs after it landed indicates a flaw in the air ventilation system in the left wing, responsible for leading air from the engines to cool the cabin. Based on the measured spectra, which indicate that the two noise sources are compact monopoles, you suspect there could be small holes in the aircraft wing frame, through which air is leaking from the ventilation tubes. To help the maintenance team inspect the wing frame, you were asked to estimate the diameters of the two holes they need to look for. In your analysis, assume the air inside the tubes is compressed to 8000 Pa above the external atmospheric pressure and neglect the air flow on the wing surface. Would the diameters you computed satisfy the underlying compactness assumption of your analysis of the noise sources? **(5 points)**

Part B - Effect of Motion on Acoustic Sources

35 points

In many practical situations, the source and/or the observer can be in motion relative to one another. For example, an aircraft flying over a stationary observer on the ground. When there is relative motion between the source and the observer, a **Doppler effect** (or Doppler shift) can occur. Named after the physicist Christian Doppler, who first described it in 1842, this effect describes the change in the observed frequency (or equivalently, the period) of a wave caused by motion between the source and the observer. If the source moves towards the observer, each successive wave cycle is emitted from a position closer to the observer than the previous one (see Fig. 3). The time between arrivals of the cycles is therefore shorter from the observer's perspective, and so the observed frequency increases. If the sound source moves away from the observer, each cycle of the wave is emitted from a position farther from the observer than the previous cycle, so the time (or period) between successive arrivals is increased, and the observed frequency decreases. A common example of Doppler shift is the change in pitch heard when a vehicle approaches and then passes an observer: the sound is heard at a higher pitch during approach, matches the emitted frequency at the instant of passing, and then drops to a lower pitch as the vehicle moves away.

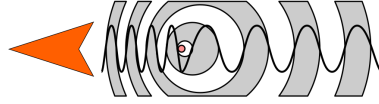


Figure 3: Change of wavelength caused by the motion of a noise source, also known as the “Doppler effect”.

Consider the general relative motion between an observer and a sound source in a Cartesian coordinate system (x_1, x_2, x_3) , as shown in Fig. 4. As shown from the schematics, a wavefront is emitted by the source at time τ , from position $\mathbf{y}(\tau)$, and is received by the observer at time t , at position $\mathbf{x}(t)$. The velocity vectors of the observer and the sound source are given by $\mathbf{v}_x$ and $\mathbf{v}_y$, respectively. The sound source emission time τ is also known as the **retarded time**. The speed of sound is c_o and $\mathbf{r}(t, \tau) = \mathbf{x}(t) - \mathbf{y}(\tau)$ is the line-of-sight vector from the source (at emission time τ) to the observer (at reception time t) with its unit vector being $\hat{\mathbf{n}}_r = \mathbf{r}/|\mathbf{r}|$. The angle θ_x is between the velocity vector $\mathbf{v}_x$ and $\hat{\mathbf{n}}_r$ and θ_y is between $\mathbf{v}_y$ and $\hat{\mathbf{n}}_r$. The fluid medium is assumed to be characterized by a uniform velocity $\mathbf{v}_o$. We restrict our general motion analysis to a subsonic observer motion, and assume: 1) observer/source motions are smooth enough to be differentiable, and 2) for each emission time τ there is a unique reception time t ($t > \tau$) along $\mathbf{r}$, known as the **causality condition**; i.e., no multi-valued mapping from τ to t for the path $\mathbf{r}$.

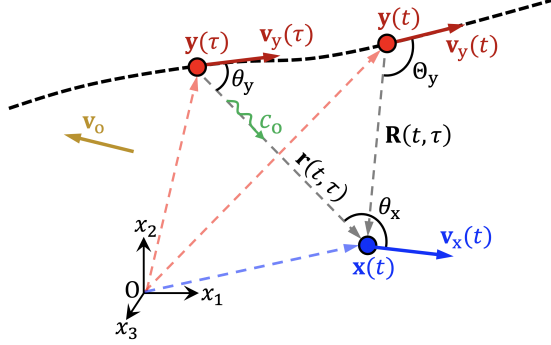


Figure 4: Schematics of a general relative motion between an observer and a sound source.

Given that the fluid medium has a uniform velocity $\mathbf{v}_o$, we need to consider two frames of reference: 1) a *stationary* ‘observer’ frame $(\mathbf{x}, \mathbf{y}, t, \tau)$, and 2) a *moving* fluid medium frame $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, t, \tau)$ in which the medium is considered at rest ($\tilde{\mathbf{v}}_o = 0$). Please note that time variables (t, τ) are considered Galilean invariant (relativity does not apply). The Galilean transformation between the two frames is:

$$\begin{aligned}\tilde{\mathbf{x}}(t) &= \mathbf{x}(t) - \mathbf{v}_o t, & \tilde{\mathbf{v}}_x(t) &= \mathbf{v}_x(t) - \mathbf{v}_o \\ \tilde{\mathbf{y}}(\tau) &= \mathbf{y}(\tau) - \mathbf{v}_o \tau, & \tilde{\mathbf{v}}_y(\tau) &= \mathbf{v}_y(\tau) - \mathbf{v}_o.\end{aligned}\quad (5)$$

In the *moving* frame, we can define the distance vector from the source to observer as: $\tilde{\mathbf{r}}(t, \tau) = \tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)$. In addition, since sound propagates at speed c_o , we can also write:

$$|\tilde{\mathbf{r}}(t, \tau)| = c_o \cdot (t - \tau) \equiv |\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|. \quad (6)$$

As a result, we can define the retarded time as follows:

$$\tau = t - |\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|/c_o. \quad (7)$$

Please notice that for the above equations to hold, a causality condition has to be satisfied, that is $t > \tau$. In other words, given some distance between the source and the observer, sound sources emitted at some time τ can only be heard by the observer at larger times. We shall now introduce the following function:

$$g(t, \tau) = t - \tau - |\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|/c_o, \quad (8)$$

where we recognize that $g(t, \tau) = 0$ to satisfy Eq. 6. Based on the causality condition ($t > \tau$), we note that any small change in the emission time $\delta\tau$ will result in a small change in the reception time δt . Thus, we can conclude that t depends on τ ; $t = t(\tau)$. Recognizing that $g(t(\tau), \tau) = 0$ and based on our previous assumptions (smooth motions and subsonic observer motion), we can differentiate g with respect to τ to yield:

$$\frac{d}{d\tau}(g(t(\tau), \tau)) = \frac{\partial g}{\partial t} \frac{dt}{d\tau} + \frac{\partial g}{\partial \tau} = 0. \quad (9)$$

By computing $\partial g/\partial t$ and $\partial g/\partial \tau$:

$$\begin{aligned}\frac{\partial g}{\partial t} &= 1 - \frac{1}{c_o} \frac{\partial}{\partial t} (|\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|) = 1 - \frac{\tilde{\mathbf{n}}_r \cdot \tilde{\mathbf{v}}_x(t)}{c_o} \\ \frac{\partial g}{\partial \tau} &= -1 - \frac{1}{c_o} \frac{\partial}{\partial \tau} (|\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|) = \frac{\tilde{\mathbf{n}}_r \cdot \tilde{\mathbf{v}}_y(\tau)}{c_o} - 1,\end{aligned}\tag{10}$$

we can extract an expression for $dt/d\tau$. Recognizing that $dt/d\tau = f_y/f_x$, where f_y is the emitted sound frequency, and f_x is the received frequency by the observer, we can write:

$$\frac{f_x}{f_y} = \frac{c_o - \tilde{\mathbf{n}}_r \cdot (\mathbf{v}_x(t) - \mathbf{v}_o)}{c_o - \tilde{\mathbf{n}}_r \cdot (\mathbf{v}_y(\tau) - \mathbf{v}_o)},\tag{11}$$

where $\tilde{\mathbf{n}}_r$ is the line-of-sight unit vector from source to observer in the *moving* frame, defined as:

$$\tilde{\mathbf{n}}_r = \frac{\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)}{|\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_o(t - \tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_o(t - \tau)|}.\tag{12}$$

In most low Mach uniform flow applications ($|\mathbf{v}_o|/c_o \ll 1$) and far-field observers ($|\mathbf{x}(t) - \mathbf{y}(\tau)| \gg 1$), we can approximate $\tilde{\mathbf{n}}_r \approx \hat{\mathbf{n}}_r$, thus simplifying Eq. 11 into:

$$\frac{f_x}{f_y} \approx \frac{c_o + |\mathbf{v}_x(t) - \mathbf{v}_o| \cos \tilde{\theta}_x}{c_o - |\mathbf{v}_y(\tau) - \mathbf{v}_o| \cos \tilde{\theta}_y} = \frac{1 + M_{x_r}(t) \cos \tilde{\theta}_x}{1 - M_{y_r}(\tau) \cos \tilde{\theta}_y},\tag{13}$$

where $M_{x_r} = |\mathbf{v}_x(t) - \mathbf{v}_o|/c_o$ and $M_{y_r} = |\mathbf{v}_y(\tau) - \mathbf{v}_o|/c_o$ are the observer and source Mach numbers relative to the flow, respectively. $\tilde{\theta}_x$ is the angle between the observer's relative velocity vector $\mathbf{v}_x(t) - \mathbf{v}_o$ and $\hat{\mathbf{n}}_r$, and $\tilde{\theta}_y$ is the angle between the relative source velocity vector $\mathbf{v}_y(\tau) - \mathbf{v}_o$ and $\hat{\mathbf{n}}_r$. Note that when $\mathbf{v}_o = 0$, $\tilde{\theta}_x = \theta_x$ and $\tilde{\theta}_y = \theta_y$.

- **B.1** Consider the following retarded time equation:

$$c_o(t - \tau) = |\mathbf{x} - \mathbf{y} - \mathbf{v}_o(t - \tau)|\tag{14}$$

You are asked to solve the equation above and define when there is a solution for the reception time t . To simplify the calculation, consider a point source located at $\mathbf{y} = 0$ and emitting a pulse at $\tau = 0$, and follow the steps below.

1. Develop the equation: $(c_o^2 - |\mathbf{v}_o|^2) t^2 + 2(\mathbf{x} \cdot \mathbf{v}_o) t - |\mathbf{x}|^2 = 0$. **(2 points)**
2. Show that there is only a real solution for t when the observer $\mathbf{x}$ is inside the Mach cone defined by $\sin \theta = 1/M$, where θ is the angle between the mean flow velocity $\mathbf{v}_o$ and the observer position $\mathbf{x}$, and $M = |\mathbf{v}_o|/c_o$ is the mean Mach number. **(6 points)**

- **B.2** A source is moving with a constant speed and a subsonic Mach number M along the x_1 -direction in a two-dimensional Cartesian coordinate system (x_1, x_2) . A stationary observer is placed on the ground beneath the source, with d being the minimum source to observer distance. The fluid medium is at rest ($\mathbf{v}_o = 0$).
 1. Plot the frequency heard by the observer as a function of time when the source is moving at $M = 0.2, 0.4$, and 0.6 . Normalize the scales on the plot as ω/ω_0 and tc_o/d , where c_o is the speed of sound, ω is the angular frequency heard by the observer at some distance r , and ω_0 is the angular frequency observed when $r = d$. Discuss the physical interpretation of the results, the effect of Mach number, and the differences between a source approaching the observer and moving away from it. **(10 points)**
 2. How your answer above would change if both the source and observer were kept stationary in a fluid medium that is characterized by a velocity vector of $\mathbf{v}_o = (v_o, 0)$, given $v_o/c_o = 0.2, 0.4$, and 0.6 . **(2 points)**
- **B.3** Commonly in aeroacoustics, we wish to determine the acoustic pressure $p'(\mathbf{x}, t)$ at observer location $\mathbf{x}$ and at time t due to the waves in the fluid and the surface sources, defined by the space coordinate $\mathbf{y}$ and time τ . To do this, we utilize the Green's function G , defined as a solution to the inhomogeneous wave equation given below:

$$\frac{1}{c_0^2} \frac{\partial^2 G}{\partial \tau^2} - \frac{\partial^2 G}{\partial y_i^2} = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau), \quad (15)$$

where the functions on the right of Eq. 15 are Dirac delta functions:

$$\delta(\mathbf{x} - \mathbf{y}) = \begin{cases} \infty, & \text{if } \mathbf{x} = \mathbf{y} \\ 0, & \text{if } \mathbf{x} \neq \mathbf{y} \end{cases} \quad \delta(t - \tau) = \begin{cases} \infty, & \text{if } t = \tau \\ 0, & \text{if } t \neq \tau \end{cases} \quad (16)$$

Note that the Green's function is a solution of the inhomogeneous wave equation in space as a function of $\mathbf{y}$ and time τ but also depends on the observer position $\mathbf{x}$ and time t , where we wish to obtain our solution and so has a functional form $G(\mathbf{x}, t | \mathbf{y}, \tau)$.

1. For a fixed observer at location $\mathbf{x}$ and fixed time t , show that the free field Green's function given by:

$$G_0(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - \tau - |\mathbf{x} - \mathbf{y}|/c_o)}{4\pi|\mathbf{x} - \mathbf{y}|} \quad (17)$$

represents a collapsing impulsive wave onto the point $\mathbf{x}$ arriving at time t when $t > \tau$ and is zero when $t < \tau$. **(5 points)**

2. For a fixed location $\mathbf{y}$ and time τ , show that the free field Green's function (see Eq. 17) represents an outgoing impulsive wave from the point $\mathbf{y}$ initiated at time τ when $t > \tau$ and is zero when $t < \tau$, and hence satisfy the wave equation (see Eq. 15) away from the source point $(\mathbf{x}, t) \neq (\mathbf{y}, \tau)$, for which:

$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G}{\partial y_i} - G \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i dS(\mathbf{y}) d\tau \quad (18)$$

represents waves propagating away from the surface S . **(10 points)**

In your answer, make sure to work in spherical coordinates and make use of these identities:

$$\nabla^2(g(r)) = \frac{\partial^2 g(r)}{\partial r^2} + \frac{2}{r} \frac{\partial g(r)}{\partial r}, \quad r = |\mathbf{x} - \mathbf{y}| \quad (19)$$

Part C - Airframe Cavity Noise

35 points (+ 15 bonus points)

Cavity flow is a fluid dynamic configuration that consists of a plate with a rectangular cutout (cavity) immersed in a fluid moving with velocity U_∞ parallel to the plate with a boundary layer thickness δ . A schematic description of the cavity flow configuration is shown in Fig. 6. The cavity dimensions are given by L (length), D (depth), and W (width). Here, the cavity is characterized by a right-handed Cartesian coordinate system (x, y, z) , with the origin at the centerline of the cavity upstream wall leading edge.

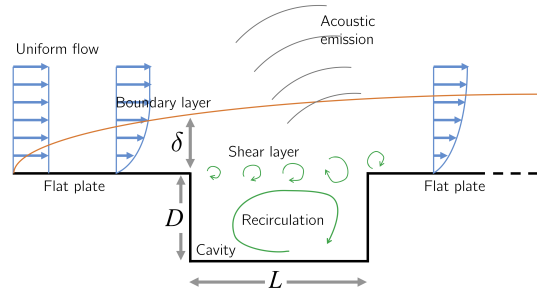


Figure 5: Illustration of the open cavity flow [2].

The sudden geometric discontinuity at the cavity (opening) causes the incoming boundary-layer flow to separate from the leading edge of the upstream wall, forming a free shear layer that spans across the cavity length and encloses

a recirculation region. The strong velocity gradient across this shear layer, between the high-speed outer flow and the low-speed recirculating flow inside the cavity, makes the shear layer inherently unstable, giving rise to the well-known Kelvin–Helmholtz instability; any small disturbances in the shear layer grow as they convect downstream in the form of vortical structures. When these shear-layer vortices impinge on the downstream wall of the cavity, their rapid deformation and compression generate upstream-propagating pressure perturbations in the form of acoustic waves. When these upstream-propagating perturbations reach the leading edge of the cavity upstream wall, they modulate the separation process and trigger the shedding of new vortices in the shear layer. These newly formed vortices are again amplified by the shear-layer instability as they convect downstream, thus closing a self-sustained hydro-acoustic feedback loop. When the phase and gain conditions are satisfied, this loop produces distinct narrow-band tones in the measured acoustic spectra (see Fig. 6).

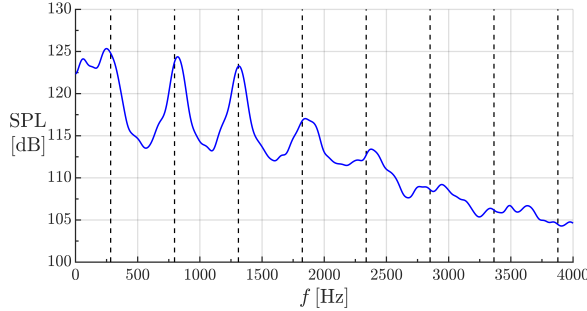


Figure 6: An example SPL spectra of the surface pressure perturbations measured at the upstream wall of a cavity that is part of a generic jet-fighter model at Mach number of $M_\infty = 0.9$ [3]. The vertical dashed lines highlight the predictions of the harmonics using Rossiter’s semi-empirical model [4].

In this section, you will gain a fundamental understanding of the flow-induced noise in cavities through the development and application of Rossiter’s model. We start by assuming a two-dimensional, semi-infinite flat plate with a fully developed boundary-layer flow, characterized by a freestream velocity U_∞ and Mach number M_∞ (see Fig. 6).

- **C.1** Can a self-sustained hydro-acoustic feedback loop take place inside the cavity if the flow entering the cavity is characterized by a laminar boundary layer? What flow phenomenon and flow parameters play a vital role in the formation of the feedback loop? Please explain your answer. **(5 points)**

John E. Rossiter [4] identified the aforementioned self-sustaining hydro-acoustic feedback loop mechanism in cavities and produced a semi-empirical equation to estimate its associated harmonics in 1964. The model assumes that upstream-propagating acoustic waves generated at the downstream wall of the cavity induce shedding of vortices at the leading edge of the cavity upstream

wall. This vorticity, in turn, is responsible for the acoustic emission as it interacts with the downstream wall. In the following question, you will derive Rossiter's semi-empirical formula for the resonant frequencies (cavity tones) of an open rectangular cavity of length L in a uniform subsonic flow ($M_\infty \ll 1$), where $M_\infty = U_\infty/c_o$.

- **C.2** Assume a one-dimensional problem in which the separated shear layer spans along the cavity length, with the vortices convecting downstream at speed $U_v = \kappa U_\infty$, where κ is a proportion coefficient. These vortices then impinge on the downstream wall, generating acoustic waves with wavelength λ_a that propagate upstream at speed U_a , exciting new vortices at the upstream wall. Start by considering a single scale of discrete vortices entering the cavity and spanning along its length, where the spacing between two vortices is the aerodynamic wavelength λ_v . Assume the acoustic radiation initiates the vortex shedding from the leading edge of the cavity upstream wall, and that the impingement of the vortices on the downstream wall is responsible for the acoustic radiation.

1. Using the flow parameters above, write an expression for the hydrodynamic vortex shedding frequency f_v and the acoustic frequency f_a . In the case of self-sustained hydro-acoustic oscillations in a cavity, should the two frequencies be equal (i.e., $f = f_a = f_v$)? **(2 points)**
2. Consider two time instants. At the initial time $t = 0$, the phase of the upstream-propagating acoustic wave is zero, so the cavity spans a complete number (n_a) of acoustic wavelengths (λ_a), and a discrete vortex is located at some given distance x_v past the downstream edge of the cavity. At a later time $t = t_0$, an upstream-propagating acoustic wavefront reaches the upstream edge of the cavity at the same time that a new vortex is shed there, where now a complete number (n_v) of vortex motion wavelengths (λ_v) are involved. Write two kinematic expressions, one for the discrete vortices and another for the acoustic wavefronts, describing their motion along the cavity length L over the time interval $[0, t_0]$. **(5 points)**
3. Eliminate the time t_0 between the two expressions you derived above and rearrange the result to obtain an equation of the form:

$$\frac{U_v}{U_a} n_a \lambda_a + (n_v - \alpha_v) \lambda_v = L \left(1 + \frac{U_v}{U_a} \right) \quad (20)$$

where α_v is a constant between 0 to 1. **(3 points)**

4. Show that the frequency of the self-sustained hydro-acoustic oscillations, known as Rossiter's semi-empirical model, can be expressed as:

$$f = \frac{U_\infty}{L} \frac{n - \alpha_v}{\frac{1}{\kappa} + M_\infty}. \quad (21)$$

Please discuss the assumptions made to obtain the expression above and the physical interpretation of α_v and κ .

(5 points)

- **C.3** Rossiter suggested $\alpha_v = 0.25$ and $\kappa = 0.57$ for cavities with $L/D \leq 4$, which give a good agreement with experiments. For shallower cavities ($4 \leq L/D \leq 10$), Rossiter adjusted the constant α_v empirically according to the following table.

L/D	4	6	8	10
α_v	0.25	0.38	0.54	0.58

Study the effect of Mach number ($0.1 \leq M_\infty \leq 0.8$) and L/D ($= 4, 6, 8, 10$) on Strouhal number, defined as $St_L = fL/U_\infty$, for the first six Rossiter modes ($n = 1, \dots, 6$). Please plot the results and discuss the effects. **(5 points)**

- **C.4** Consider an aircraft that approaches to land along a straight path $\mathbf{y}_1$ parallel to the ground at an altitude of $h = 500$ ft and a constant airspeed of $M_\infty = 0.2$ (TAS), as shown in Fig. 7. During this approach, a total of three landing gears are deployed (one at the nose and two main landing gears close to the wing root), each from its rectangular bay (cavity). Assume each landing gear consists of 0.52 m diameter wheels and a strut assembly with a 0.18 m diameter, and that the cavity has dimensions of $L = 1$ m and $D = 0.5$ m. The principal source of noise from the landing gear elements is the unsteady side force resulting from vortex shedding, for which the fundamental frequency $St = fd/U_\infty \approx 0.2$, where U_∞ is the flow velocity and d the component diameter. Assume the speed of sound is $c_o = 343$ m/s. In addition, each bay element exhibits four dominant Rossiter modes. Neglect any aerodynamic or acoustic interactions between the landing gears and bays. Assume the landing gears and bays are subjected to the same freestream velocity.

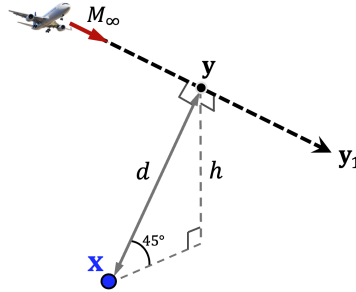


Figure 7: Schematics of the aircraft approaching for landing along the path x_1 , assumed to be parallel to the ground.

1. Determine whether, for a landing Mach number of $M_\infty = 0.2$, the wheels and strut assembly can be considered acoustically compact sources? Would the bays be considered as acoustically compact sources? If so, please specify what Rossiter modes satisfy the compactness criteria. **(2 points)**

2. A stationary ground microphone was placed at $\mathbf{x}$ beneath the aircraft flying path $\mathbf{y}_1$ at 45° to the shortest distance d to $\mathbf{y}_1$ (see Fig. 7). For both the landing gear elements and bays, would they be considered in the acoustic far field if measured at the microphone location $\mathbf{x}$? **(2 points)**
3. The ground microphone measured the acoustic signal during an aircraft flyover test along the flight path $\mathbf{y}_1$. The measurement window was $[-t_0, t_0]$, assuming the time the aircraft passed the point $\mathbf{y}$ is $t = 0$. Acoustic spectra results showed two dominant harmonics during an aircraft flyover scenario. Estimate the two narrow-band frequencies measured during the flyover maneuver. Please discuss the noise sources associated with each frequency. **(6 points)**
4. **Bonus** Please estimate the minimum measurement window $2t_0$ and minimum sampling rate needed to accurately resolve the narrow-band spectral content measured by the ground microphone. **(3 points)**
5. **Bonus** The unsteady force on each strut and wheel assembly (directed in the spanwise direction) has an RMS amplitude of $2F_{\text{rms}}/\rho_o U_\infty^2 S = 0.2$, where $\rho_o = 1.225 \text{ kg/m}^3$ is the ambient air density and $S_{\text{strut}} = 0.1 \text{ m}^2$ and $S_{\text{wheel}} = 0.2 \text{ m}^2$ are the effective area of the strut and wheel assembly, respectively. Estimate the SPL in dB(re $p_{\text{ref}} = 20 \mu\text{Pa}$) at the ground microphone due to the strut assemblies, as well as due to the wheel assemblies, when the aircraft is at the location $\mathbf{y}$. What are the frequencies associated with the SPL values you computed? Ignore scattering from any other parts of the aircraft and assume convection effects on the propagation of sound are negligible. You can also assume all the wheel elements are in-phase, as well as the strut elements. **(6 points)**
6. **Bonus** During the aircraft flyover, each bay was instrumented with a unsteady force transducer, measuring an RMS force amplitude (directed in the normal direction) of $2F_{\text{rms}}/\rho_o U_\infty^2 S_{\text{cavity}} = 0.1$, where $S_{\text{cavity}} = 2 \text{ m}^2$ is the effective area of the bay walls. Estimate the SPL in dB(re $p_{\text{ref}} = 20 \mu\text{Pa}$) of the dominant cavity frequency at the ground microphone due to the cavity elements, when the aircraft is at the location $\mathbf{y}$. Ignore scattering from any other parts of the aircraft and assume convection effects on the propagation of sound are negligible. You can also assume that all the cavity elements are in phase. Discuss the meaning of the results you computed. What can we learn about the two narrow-band frequencies identified in the measured acoustic spectra during the aircraft flyover test? **(6 points)**

References

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- [2] Marlon Sproesser Mathias and Marcello A. F. Medeiros. The effect of incoming boundary layer thickness and Mach number on linear and nonlinear Rossiter modes in open cavity flows. *Theoretical and Computational Fluid Dynamics*, 35(4):495–513, 2021.
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